Usability Case Study

Amazon's Checkout Flow: A Heuristic Evaluation

Sujan Pokhrel

DGM 6168 — Spring 2026

Framework: Nielsen's 10 Usability Heuristics

1. Product Context and User Understanding

Before evaluating, we need to be clear about what Amazon's checkout flow is, who uses it, and what a successful interaction looks like.

What is the product?

Amazon's checkout flow is the multi-step path a logged-in shopper takes from **cart** through **shipping address**, **payment method**, **order review**, and finally **order confirmation**. It is the highest-value transactional surface in e-commerce — Amazon processed over $574 billion in net sales in 2023 (Amazon, 2024), nearly all of which flowed through these screens. This evaluation focuses on the desktop web browser version of the flow, accessed at amazon.com by an authenticated shopper purchasing a single physical item.

Who are the users?

The user base is enormous and heterogeneous, but for this evaluation we focus on the primary persona that drives the design: repeat shoppers placing routine physical-goods orders. The relevant characteristics include:

- Age range 18-65, with a large concentration in 25-54 (Statista, 2024).
- Moderate technical familiarity — comfortable with online forms but not necessarily attentive readers.

- Mixed devices, with the desktop flow used heavily for higher-value or work-related purchases.
- Task-focused goal orientation. Most users want to finish checkout and move on, not browse.
- Generally high trust in Amazon, which is both an asset and a vulnerability — users may click through confirmations without reading.

What does success look like?

For the user, a successful checkout is one where they complete the purchase quickly, see exactly what they will be charged before they commit, do not accidentally add unwanted items or subscriptions, and can correct mistakes easily after submission. For Amazon, success is measured in completion rate, average order value, and reduction in returns and cancellations.

These two definitions of success are not always aligned, and the tensions between them surface as several of the usability issues identified in this report. When the user's goal (clarity, control) competes with the business goal (upselling, subscription enrollment), the design choices that resolve the conflict reveal the product's true priorities.

Scope of this evaluation

The evaluation covers the four screens of the live checkout in sequence: Shopping Cart; Secure Checkout, which combines the delivery address, payment method, and an order-review stub on a single page (and includes an error state); Review and Place Order; and Order Confirmation. Mobile-app and Buy-Now one-click flows are out of scope, as are guest checkout and digital-goods purchases. The framework applied is Jakob Nielsen's 10 Usability Heuristics (Nielsen, 1994a) with severity ratings on Nielsen's 0-4 scale (Nielsen, 1994b).

2. Methodology

A heuristic evaluation is a usability inspection method in which evaluators examine an interface against a recognized set of design principles, called heuristics, and identify violations (Nielsen, 1994a). This study follows the standard three-phase protocol described by Nielsen (1994b) and by Moran and Gordon (2024):

Procedure

Walkthrough. The evaluator performed the full checkout task on the live site — adding an item to the cart and proceeding through to the order-review step — capturing an annotated screenshot at each stage.

Issue identification. Each screen was reviewed against all 10 of Nielsen's heuristics. Any element that conflicted with one or more heuristics was logged as a potential issue.

Severity rating. Each confirmed issue was rated using Nielsen's 0-4 severity scale, weighing frequency, impact, and persistence (Nielsen, 1994b).

Nielsen's 10 heuristics, applied to each screen:

1. Visibility of system status — keep users informed about what is going on.
2. Match between system and the real world — speak the user's language.

3. User control and freedom — provide clearly marked exits and undo.
4. Consistency and standards — follow conventions; don't make users wonder.
5. Error prevention — design out errors before they happen.
6. Recognition rather than recall — make options visible; minimize memory load.
7. Flexibility and efficiency of use — let experts work fast without breaking it for novices.
8. Aesthetic and minimalist design — strip irrelevant information out.
9. Help users recognize, diagnose, and recover from errors — plain language, precise problems, constructive solutions.
10. Help and documentation — concise, task-focused, easily searchable.

Severity scale

Nielsen's standard 0-4 severity scale is used throughout this report:

- 0 — Not a usability problem.
- 1 — Cosmetic only; fix if time permits.
- 2 — Minor; low priority fix.

- 3 — Major; important to fix; high priority.
- 4 — Catastrophe; imperative to fix before release.

Severity ratings here are this evaluator's judgments based on the criteria above. In a multi-evaluator study they would be averaged across evaluators per Nielsen's recommendation; for this single-evaluator case study they should be read as one informed perspective.

Note on screenshots

Screenshots were captured from the live Amazon.ca checkout on June 15, 2026; personal data has been redacted and replaced with placeholders, with nothing else altered.

3. Findings

This section presents the eight usability issues identified across the four checkout screens, organized by screen. Each finding names the violated heuristic(s), assigns a severity rating, and explains the impact on users.

Screen 1: Shopping Cart

The cart sits at the top of the funnel. Friction here costs more abandonment than friction anywhere else in the flow. Three issues stand out.

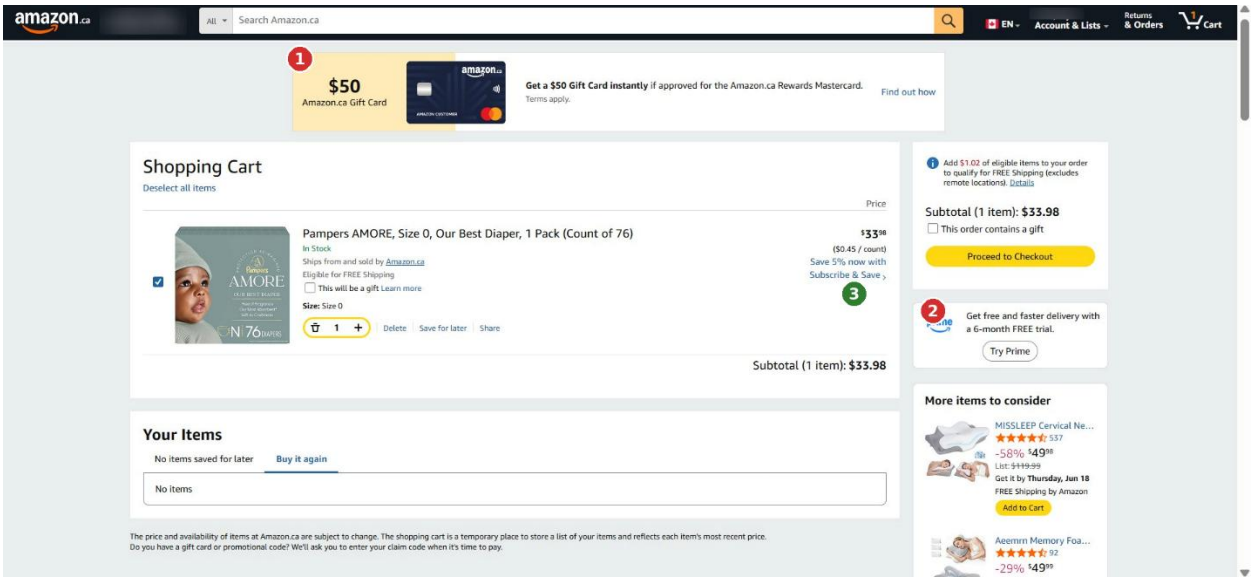


Figure 1. The Amazon.ca shopping cart with a single item, annotated with two minor issues and one positive example.

Issue 1: A credit card offer sits above the items in the cart Heuristics violated: H8 (Aesthetic and minimalist design), H1 (Visibility of system status) Severity: 2 — Minor
The first thing on the cart page isn't your item — it's a full-color banner offering a $50 gift card if you're approved for the Amazon.ca Rewards Mastercard. The product you actually came for sits underneath it.

Nielsen's eighth heuristic is about keeping a screen focused on what's relevant, since every extra block of content competes for attention with the content that matters (Nielsen, 1994a), and here the single most prominent thing on a checkout page is a credit card application. It also pushes the real cart contents further down, so the shopper has to look past a financial promotion to confirm what they're buying, which slightly muddies their sense of where they are. I've kept this minor rather than major because the banner is easy to scroll past — it adds clutter and a moment of friction, but it doesn't block the task or mislead anyone into a commitment.

Issue 2: A Prime trial offer shares space with the checkout button
Heuristics violated: H8 (Aesthetic and minimalist design) Severity: 2 — Minor
In the right-hand column, just below the subtotal and the yellow "Proceed to Checkout" button, there's a panel inviting the user to start a six-month free Prime trial. That spot, right beside the main action, is the most-looked-at part of the page, and instead of keeping attention on checkout it puts a second thing in front of the user to click. To Amazon's credit, the offer is clearly labeled and nothing is pre-selected, so this isn't a deceptive pattern — just a distraction placed at the worst possible moment. The same minimalist principle applies (Nielsen, 1994a): the checkout area should be working to get the shopper out the door, not surfacing a subscription decision. Minor, for the same reason as Issue 1 — friction, not derailment.

Issue 3: Subscribe & Save is handled the right way (a positive example)
Heuristics: H4 (Consistency and standards) — satisfied Severity: 0 — Not a usability problem
Not everything here is a problem, and it's worth saying so directly. The Subscribe & Save option, which enrolls the user in recurring deliveries, shows up as a small blue text link under the price ("Save 5% now with Subscribe & Save"), clearly separated from the yellow checkout button.

That's exactly how it should be done: a recurring-commitment option that looks like an option, not like the safe next step. It would have been easy to style it as a yellow button matching the primary action — the kind of dark pattern that turns up on plenty of other sites — and Amazon didn't. I'm including it not to pad the list but because an evaluation is more trustworthy when it notes what works alongside what doesn't; it shows the severity ratings come from what's on the screen, not from a decision to find fault everywhere.

Screen 2: Secure Checkout (delivery, payment, and review on one page)

On the live site, what the original report treated as separate "address" and "payment" screens is actually one accordion page — Delivering to → Payment method → Review items and shipping — stacked under a single "Secure checkout" header. Evaluated honestly, this is a largely well-built screen: it strips away distraction at the commit stage and defers the actual purchase correctly. What's left to criticize is minor and specific. Two issues are worth flagging, and three things the screen does well are worth recording as positive controls — including one that directly retires a finding from the earlier draft.

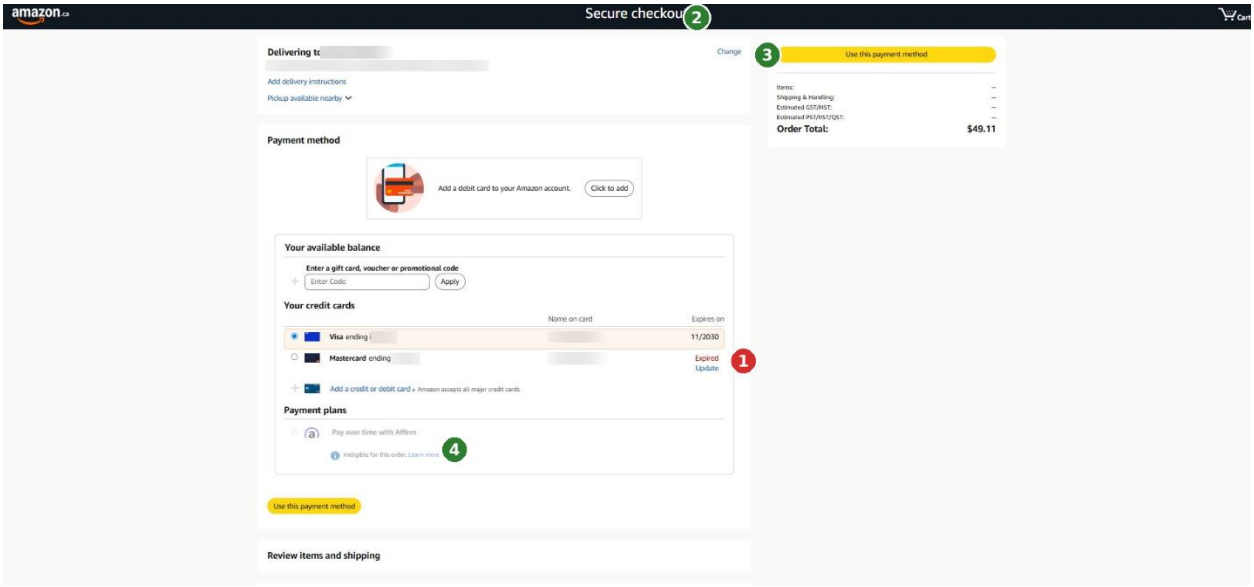


Figure 2. The single-page secure checkout, annotated with one minor issue (red) and three positive controls (green). Personal details and card numbers have been redacted.

Issue: An expired card sits in the active payment list (callout 1)
Heuristics violated: H5 (Error prevention) Severity: 2 — Minor
The "Your credit cards" list shows an expired Mastercard (marked "Expired / Update" in red) directly beneath a valid Visa, both rendered as selectable rows with radio buttons. Keeping an unusable card among the live choices — rather than moving it to a separate "needs updating"

area or dropping it from the list — invites an error path and adds a decision the user shouldn't have to make while paying. Nielsen's fifth heuristic is about designing the error out before it can happen (Nielsen, 1994a); presenting an expired option alongside valid ones does the opposite. It stays minor because the row is clearly labelled "Expired," so an attentive user won't pick it — the cost is clutter and a small cognitive tax, not a likely failure.

Issue: Two competing entry points for adding a card Heuristics violated: H8 (Aesthetic and minimalist design) Severity: 1 — Cosmetic
The Payment method section opens with a full-color promo box — "Add a debit card to your Amazon account / Click to add" — and then offers a second, separate "Add a credit or debit card" link lower in the same section. Two routes to the same action, stacked close together, is mild redundancy: it adds a unit of visual content that competes with the relevant ones (Nielsen, 1994a) and makes the user pause to work out whether the two links do different things. Cosmetic rather than minor because it causes a flicker of hesitation at most, not an error or a blocked task.

Positive control: The header is stripped to the essentials (callout 2)
Heuristics: H8 (Aesthetic and minimalist design) — satisfied Severity: 0 — Not a usability problem
A direct contrast to Screen 1: the global navigation, search bar, and recommendations are all gone, replaced by a single "Secure checkout" header. This is exactly what a commit-stage screen should do — remove everything that competes with finishing the task. The same minimalist principle the cart violated is applied correctly here.
Positive control: The primary button defers the commit (callout 3)
Heuristics: H5 (Error prevention) — satisfied Severity: 0 — Not a usability problem
The primary action reads "Use this payment method," not "Place your order." The purchase is correctly deferred to a later step, and the

button here only advances the flow. This is worth stating plainly because an earlier version of this evaluation flagged a premature "Place your order" button on the pre-payment step as a catastrophe — the live site does not behave that way. It already uses a step-appropriate CTA.

Positive control: An unavailable option explains itself (callout 4)
Heuristics: H9 (Help users recognize, diagnose, and recover from errors) — satisfied Severity: 0 — Not a usability problem
The Affirm "Pay over time" option is greyed out with "Ineligible for this order. Learn more," telling the user why it can't be used and offering a way to find out more, rather than failing silently or disappearing. Surfacing the reason for an unavailable choice is good practice.

Screen 3: Review and Place Order

This is the final screen before commit, reached after confirming the payment method. Like the payment step, it holds up well — and it specifically dismantles the two most dramatic findings in the original draft. The order is now fully itemized, the optional add-ons are all opt-in rather than pre-selected, and the contract and returns terms are spelled out. One genuine minor issue remains.

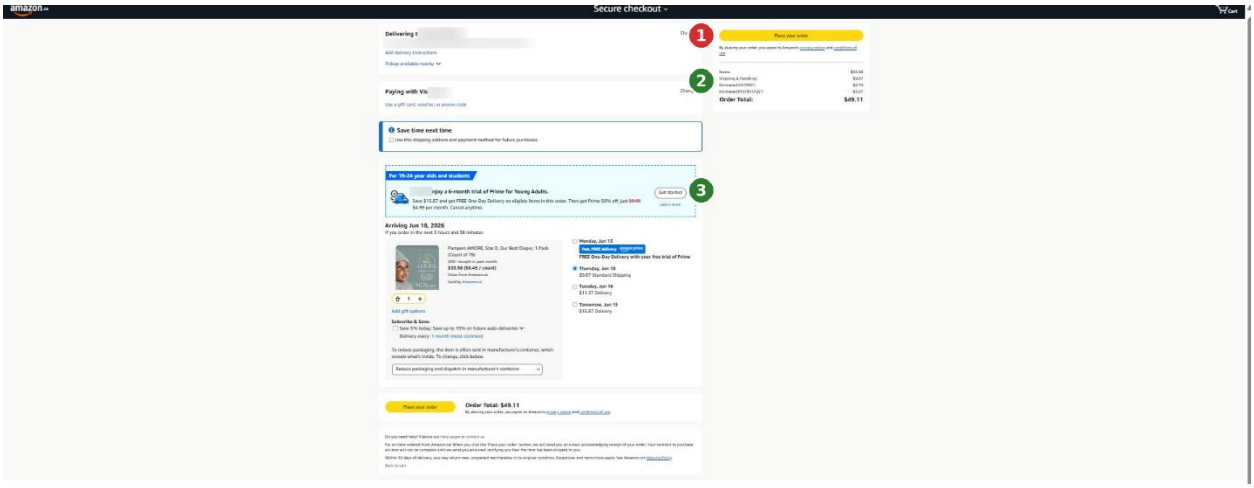


Figure 3. The review/place-order screen, annotated with one minor issue (red) and two positive controls (green). Personal details and card number redacted.

Issue 1 (callout 1): The final commit is a single click with no confirmation step Heuristics violated: H5 (Error prevention), H3 (User control and freedom) Severity: 2 — Minor Pressing "Place your order" submits the purchase immediately — there's no "Are you sure?" confirmation and no review window after the click. For a money-moving action, error-prevention principles favour confirming before an irreversible operation (Nielsen, 1994a). Two things keep this minor rather than major. First, this *is* a dedicated review screen — the user has just been shown the address, payment, items, and an itemized total, so the click isn't unconsidered. Second, the action isn't truly irreversible: the footer explains the order can be cancelled before it ships and returned within 30 days. The earlier draft rated this major and leaned on irreversibility; on the live site the order is recoverable, so

the severity is lower. The real gap is that the recovery path isn't surfaced here — which connects to the cancel-order finding on the confirmation screen.

Positive control (callout 2): The order total is fully itemized Heuristics: H1 (Visibility of system status) — satisfied Severity: 0 The summary breaks the charge down completely: Items $33.98, Shipping & Handling $9.87, Estimated GST/HST $2.19, Estimated PST/RST/QST $5.07, Order Total $49.11. Before committing, the user can see exactly what they're paying and why. This also answers a question raised on the payment screen, where these lines read "--": the breakdown simply populates by this step rather than being withheld.
Positive control (callout 3): Add-ons are opt-in, not pre-selected Heuristics: H5 (Error prevention), H3 (User control and freedom) — satisfied Severity: 0 The Prime trial is presented as a "Get started" button the user must actively click — not a pre-ticked checkbox that enrolls them by default. The "Save time next time" and Subscribe & Save options on this screen are likewise unchecked. This is the single most important correction to the original report, which flagged a pre-checked Prime opt-in as a catastrophe. The live site does the opposite — the user isn't enrolled in anything they didn't choose — so that finding has been retired.

Screen 2 (error state): Adding a card with invalid input

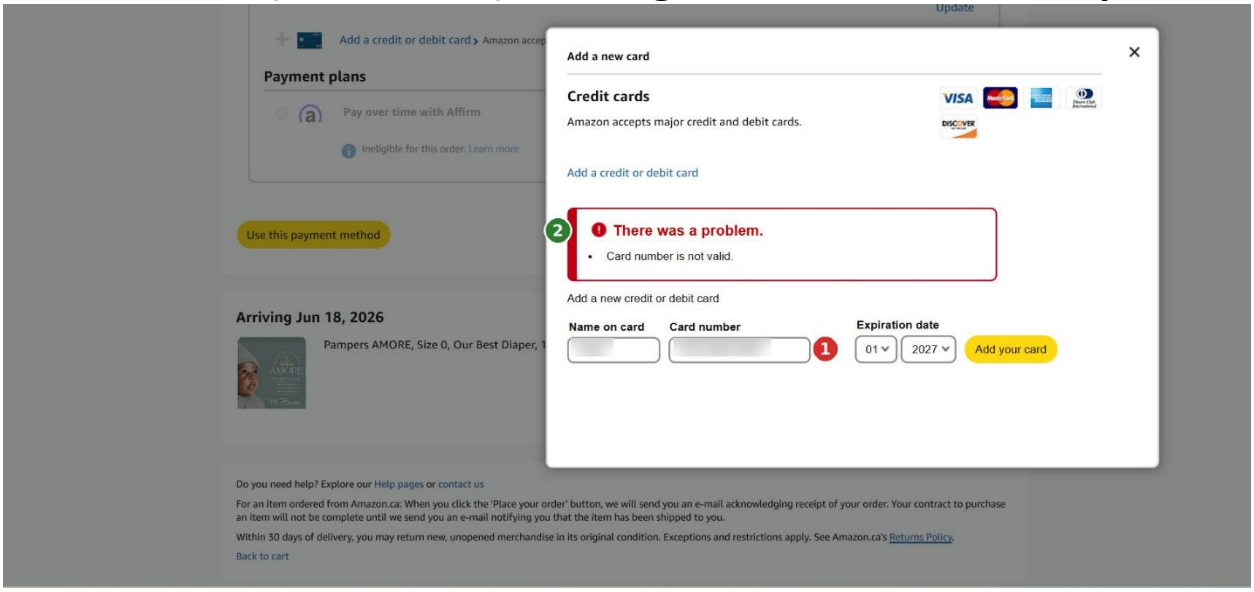


Figure 2b. The add-a-card error dialog after an invalid card number, annotated with one minor issue (red) and one positive control (green). The name and card number entered have been redacted.

Issue (callout 1): The error names the problem but not the fix, and doesn't flag the field Heuristics violated: H9 (Help users recognize, diagnose, and recover from errors) Severity: 2 — Minor The message correctly identifies which field is wrong ("Card number is not valid"), but it stops there: it doesn't suggest a constructive next step (for example, "check for a typo or try a different card"), and the invalid field itself isn't flagged inline — the error sits in a separate banner above the form while the Card number box looks unchanged. Nielsen's ninth heuristic asks for plain language, a precise problem, *and* a constructive solution (Nielsen, 1994a); this delivers the first two but not the third, and without inline highlighting the user has to map the banner back to the right field themselves. Minor, because the diagnosis is specific enough that most users will know what to do.

Positive control (callout 2): The validation error is specific and code-free Heuristics: H9 (Help users recognize, diagnose, and recover from errors) — satisfied Severity: 0 The error reads "There was a problem. Card

number is not valid" — plain language, naming the exact field, with no internal error code exposed. This is the direct correction to the original report's vague "There was a problem… Code: ERR_CC_03" with no diagnosis. The live error does the opposite, so that finding has been retired.

Screen 4: Order Confirmation (not captured)

The confirmation page is the one screen this evaluation does not capture directly, because reaching it requires placing a real, paid order. The finding below is therefore characterised from Amazon's documented post-order behaviour rather than a screenshot, and is labelled as such.

Issue: No visible cancel option immediately after the order is placed

Heuristics violated: H3 (User control and freedom) **Severity:** 3 — Major. After commit, Amazon does allow an order to be cancelled while it is still in pre-shipment status, but the confirmation page does not surface a direct "Cancel this order" control; the user must navigate to "Your Orders" and locate the order to cancel it. Nielsen's third heuristic calls for a clearly marked emergency exit (Nielsen, 1994a); the recovery path exists but is buried at exactly the moment a user who has just made a mistaken purchase needs it most. This is the most consequential of the issues found — hence the only major rating — and it is the gap Recommendation 3 closes with a visible cancel link on the confirmation page. (Based on documented behaviour, not a captured screen.)

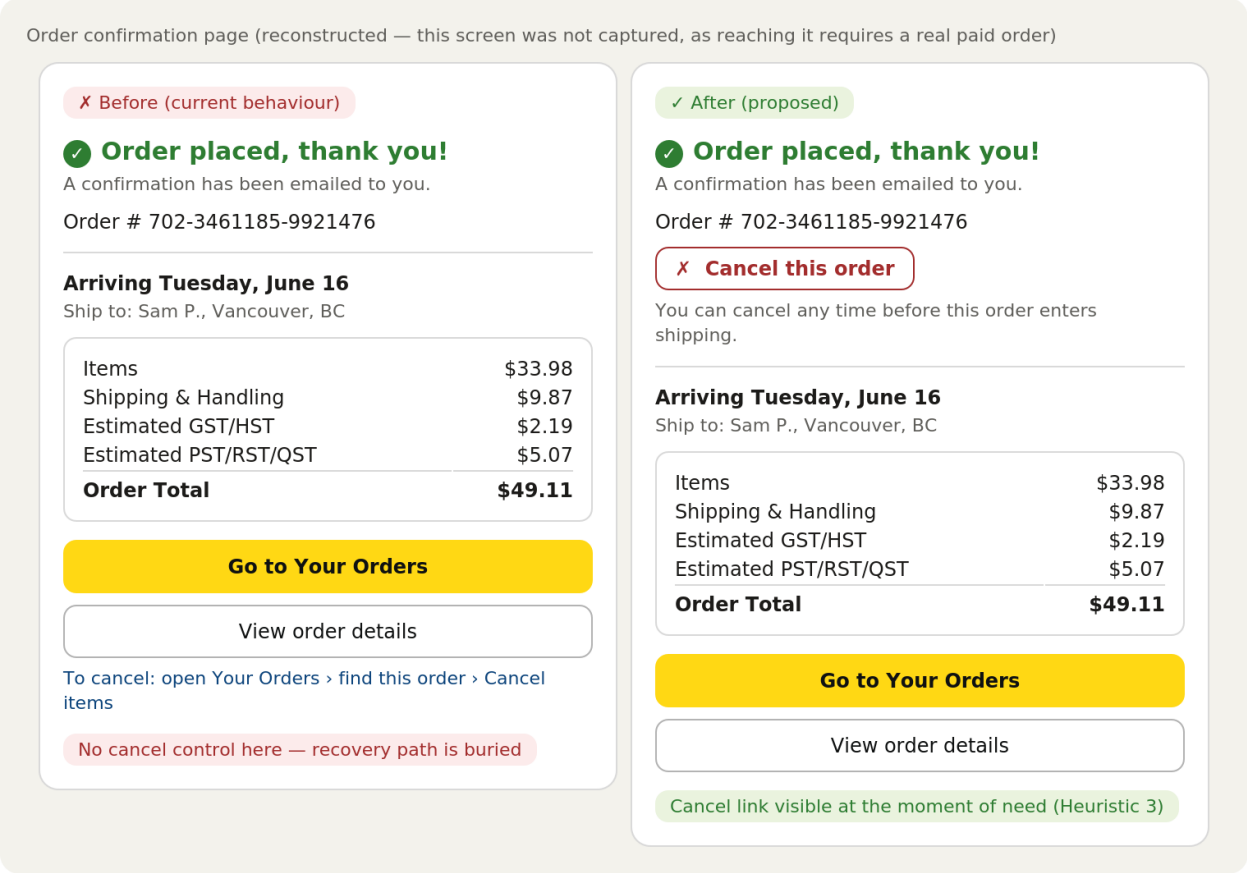


Figure 7. Order confirmation page, reconstructed (this screen was not captured). Left: the current behaviour, where the only route to cancel is buried in Your Orders. Right: the proposed fix — a visible "Cancel this order" link directly below the order number — Heuristic 3 (user control and freedom).

Additional finding (Screen 1): "Eligible for FREE Shipping" on the item, but shipping is charged at checkout Heuristics violated: H1 (Visibility of system status), H4 (Consistency and standards) **Severity:** 2 — Minor
The cart item displays "Eligible for FREE Shipping" in green directly beneath the product (Figure 1), yet the review screen charges $9.87 in Shipping & Handling (Figure 3). The two screens tell the user different things about the same cost. The phrase is technically accurate in Amazon's terms — the item is *eligible to count toward* free shipping once an order clears the threshold — but a shopper can reasonably read "Eligible for FREE Shipping" as "this ships free," and the actual

charge doesn't surface until the final review step. Visibility of system status (Nielsen, 1994a) asks that the true cost be clear up front, and the prominent green label works against that. The reason this stays minor rather than major is that Amazon *does* disclose the condition — the cart sidebar reads "Add $1.02 of eligible items to qualify for FREE Shipping" (Figure 1) — so the information is present, just easy to miss against the more eye-catching "FREE Shipping" framing. (A severity 3 is arguable because the mismatch concerns the price paid; I'd hold it at 2 given the explicit threshold disclosure.)

4. Design Recommendations

This section proposes three actionable changes, each addressing multiple findings, each grounded in a named heuristic. The recommendations are ordered by expected impact relative to implementation cost.

Recommendation 1 — Subordinate promotional content to the task at the cart and payment steps.

Addresses the cart Mastercard banner, the Prime panel, and the duplicate add-card links — Heuristic 8 (aesthetic and minimalist design). The cart and payment screens carry several elements that compete with completing the order. Three specific changes would help. Move the Amazon.ca Rewards Mastercard promotion out of the lead position above the cart items into a lower or smaller slot, so the user's product — not a credit-card application — heads the page. Consolidate the two separate "add a card" entry points on the payment screen into one clearly labelled action. And give the Prime free-trial panel secondary visual weight relative to the "Proceed to Checkout" button. Nielsen's eighth heuristic holds that every extra unit of content competes with the relevant ones (Nielsen, 1994a). Each change keeps Amazon's offer in place while stopping it from crowding the primary

action: nothing is removed, the offers are subordinated, which preserves the business goal while reducing friction.

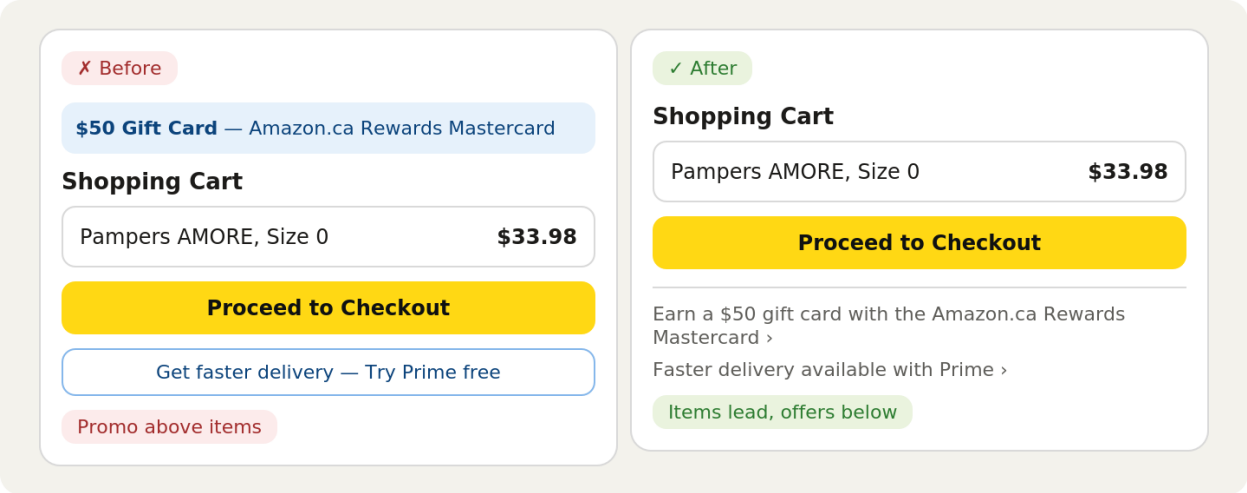


Figure 4. Recommendation 1 (proposed): the cart with promotional content subordinated — items and checkout lead, the Mastercard offer and Prime demoted to secondary links, and the duplicate add-card entry consolidated. Heuristic 8 (aesthetic and minimalist design).

Recommendation 2 — Give complete, accurate information at the point the user needs it.

Addresses the FREE-shipping mismatch and the error message without a next step — Heuristics 1 (visibility of system status), 4 (consistency and standards), and 9 (help users recover from errors).

Two places show the user only partial information. First, the cart labels the item "Eligible for FREE Shipping" while the order is actually charged $9.87, with the qualifying condition buried in a sidebar note. Replace the generic label with the real status for this order — for example, "Shipping $9.87 — add $1.02 of eligible items to ship free" — and carry

an estimated shipping line into the cart subtotal so the final total is not a surprise at review. Amazon already computes the threshold gap, so surfacing the resulting cost is low-effort, and it removes the contradiction between the cart's promise and the review's charge (Heuristics 1 and 4). Second, the card-validation error correctly says "Card number is not valid" but offers no next step and does not highlight the field. Add a brief constructive cue and inline highlighting — "Card number is not valid. Check for a typo or try a different card." with the field outlined in red — which completes the one part of Heuristic 9 the current message misses (Nielsen, 1994a).

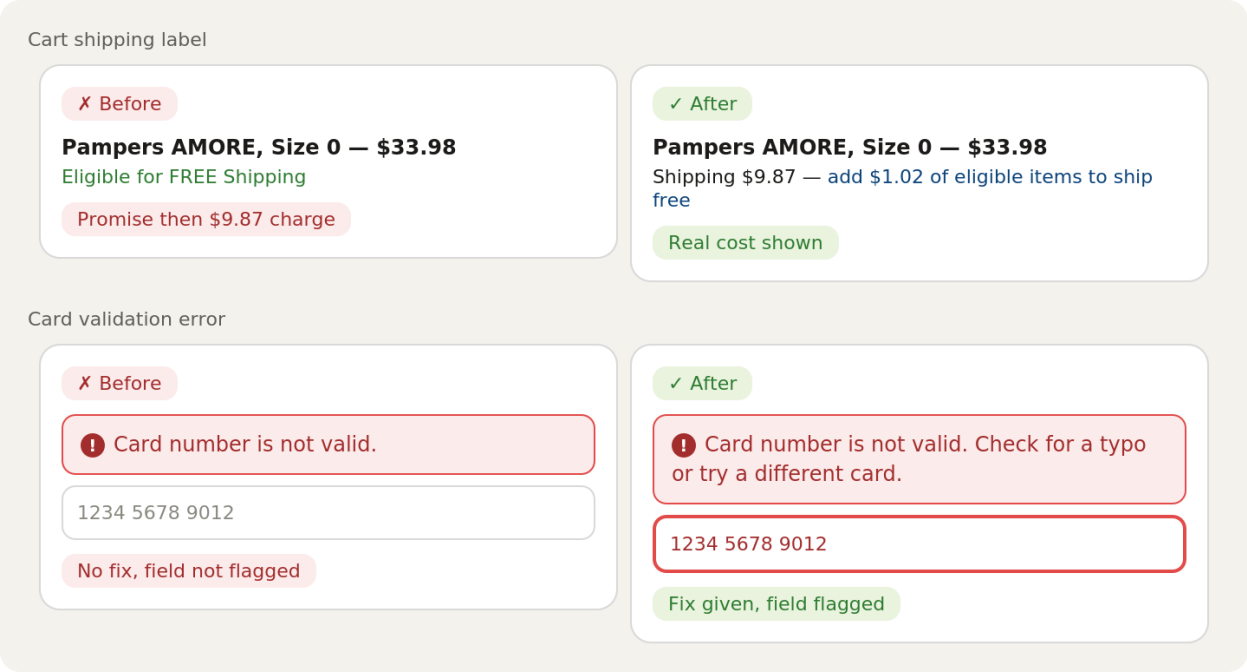


Figure 5. Recommendation 2 (proposed): the real shipping cost shown in the cart in place of the ambiguous "Eligible for FREE Shipping" label, and the card error given a constructive next step with the field

highlighted inline. Heuristics 1 (visibility of system status), 4 (consistency and standards), and 9 (help users recover from errors).

Recommendation 3 — Strengthen error prevention and user control around the commit.

Addresses the expired card, the single-click commit, and the buried cancel link — Heuristics 5 (error prevention) and 3 (user control and freedom). Three changes cluster around the moment of purchase and just after it. Remove expired cards from the active payment-selection list, or disable and visually separate them, so an unusable option cannot be chosen by mistake (Heuristic 5). For the commit itself, the single-click "Place your order" is defensible on a dedicated review screen, but for high-value orders a lightweight confirmation — or simply stating in the button area that the order can be cancelled before it ships — would prevent accidental commits at low cost (Heuristic 5). And on the confirmation page, add a visible "Cancel this order" link directly below the order number, available while the order is in pre-shipment status, rather than routing the user through "Go to your orders" to find it (Heuristic 3). Together these give the user a clearer safety net at the highest-stakes step without adding friction to the common case.

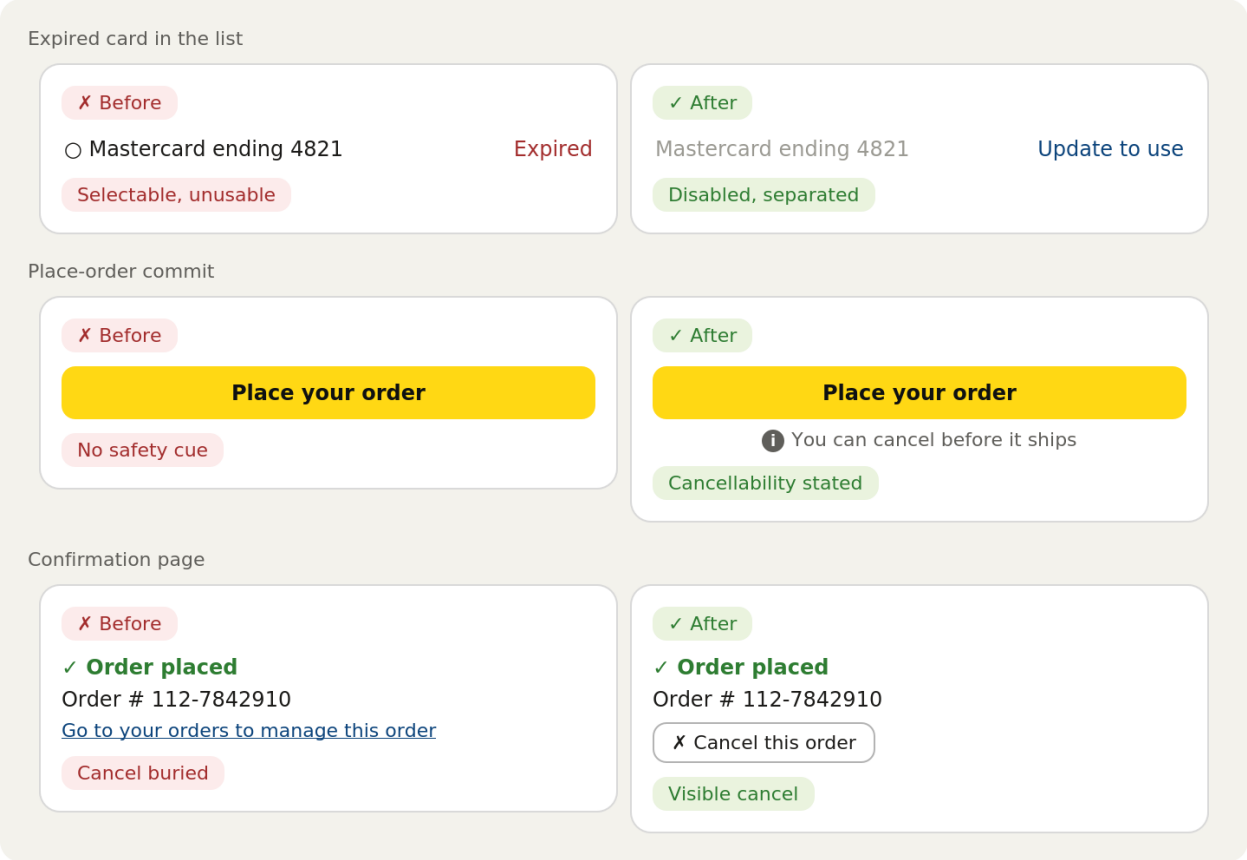


Figure 6. Recommendation 3 (proposed): expired cards disabled and separated from valid ones, a cancellability cue at the commit button, and a visible cancel link on the confirmation page. Heuristics 5 (error prevention) and 3 (user control and freedom).

What this evaluation does not address. A complete picture would still need quantitative usability testing with representative users to validate these inspection findings, a full WCAG 2.1 audit of the checkout (World Wide Web Consortium, 2018), and A/B testing of the recommendations against current conversion and abandonment metrics. A heuristic

evaluation finds issues quickly and cheaply, but it is not a substitute for testing with real users (Moran and Gordon, 2024).

Summary of findings

Across the checkout flow, this evaluation identified eight issues. Seven were observed directly on the live site; the eighth — the absence of a direct cancel option after commit — is characterised from Amazon's documented order-management behaviour rather than a captured screen, and is labelled as such in the findings. None of the eight rates as a catastrophe (severity 4). One is major (severity 3): the post-commit cancellation path, which exists but is buried. Six are minor (severity 2): the credit-card banner above the cart, the Prime panel beside checkout, the "Eligible for FREE Shipping" label that is contradicted by a shipping charge, the expired card left in the active payment list, the card-validation error that names the problem without suggesting a fix, and the single-click commit with no confirmation. One is cosmetic (severity 1): the duplicated card-entry links on the payment screen. The violations cluster under Heuristic 8 (aesthetic and minimalist design), which accounts for three of the eight — all of them promotional content layered onto the transactional path. Heuristics 1 (visibility of system status), 3 (user control and freedom), and 5 (error prevention) account for two each, and Heuristic 9 (help users recognize, diagnose, and recover from errors) for one. This pattern is worth noting: rather than the broken error-handling or pre-selected opt-ins that checkout critiques often target, the live checkout's weaknesses are concentrated in upselling and promotional density, while its high-stakes mechanics — the commit action, the opt-ins, and the validation error — are handled comparatively well. That reading is supported by the positive controls recorded alongside the issues: the stripped-down "Secure checkout" header, the step-appropriate calls to action, the fully itemized order total, the opt-ins that are unchecked by default, and the specific, code-free error message. Recording these confirms the

severity ratings reflect what is on the screen rather than a predisposition to find fault. The promotional friction that does remain is consistent with the Baymard Institute's finding that a checkout perceived as long or complicated is itself a meaningful driver of abandonment (Baymard Institute, 2023).

5. References

References are formatted in APA 7th edition style. All URLs were accessed in Spring 2026.

Amazon.com, Inc. (2024). *Amazon.com, Inc. annual report 2023*. U.S. Securities and Exchange Commission. https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK=0001018724

Baymard Institute. (2023). *Checkout usability: Research-based UX benchmark*. https://baymard.com/checkout-usability

Moran, K., & Gordon, K. (2024). *How to conduct a heuristic evaluation*. Nielsen Norman Group. https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/

Nielsen, J. (1994a). *10 usability heuristics for user interface design*. Nielsen Norman Group. https://www.nngroup.com/articles/ten-usability-heuristics/

Nielsen, J. (1994b). *Severity ratings for usability problems*. Nielsen Norman Group. https://www.nngroup.com/articles/how-to-rate-the-severity-of-usability-problems/

Statista. (2024). *Distribution of Amazon.com users in the United States as of 2024, by age group*. https://www.statista.com/statistics/amazon-user-demographics/

World Wide Web Consortium. (2018). *Web content accessibility guidelines (WCAG) 2.1*. https://www.w3.org/TR/WCAG21/